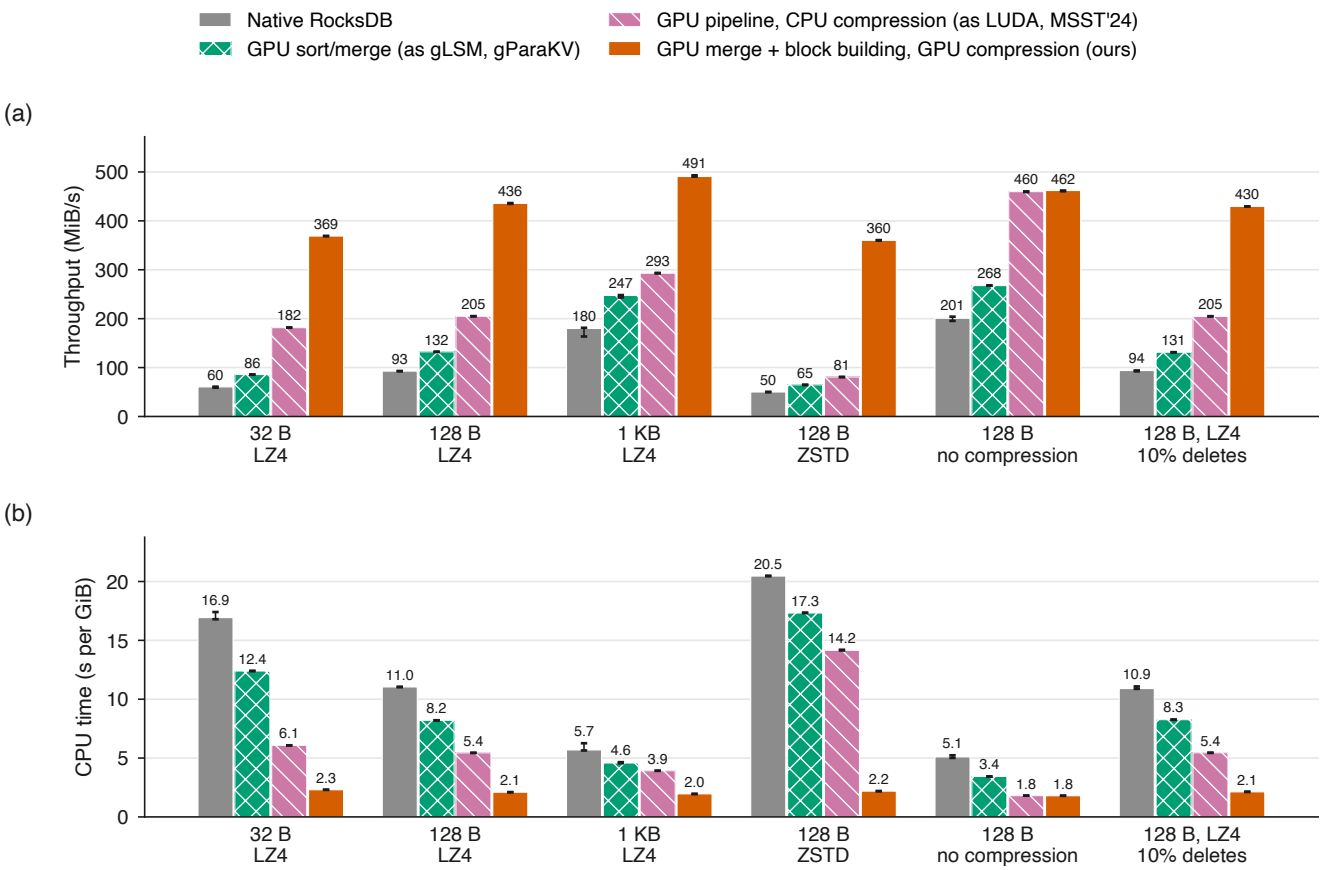
Native RocksDB
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24)
GPU merge + block building, GPU compression (ours)
(a)
Throughput (MiB/s)
0
100
200
300
400
500
60
86
182
369
93
132
205
436
180
247
293
491
50
65
81
360
201
268
460
462
94
131
205
430
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes
(b)
CPU time (s per GiB)
0
5
10
15
20
16.9
12.4
6.1
2.3
11.0
8.2
5.4
2.1
5.7
4.6
3.9
2.0
20.5
17.3
14.2
2.2
5.1
3.4
1.8
1.8
10.9
8.3
5.4
2.1
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes